

Joshua (Stapley) Montoya

West Hollywood, CA | Portfolio: <https://josh-monto.github.io/>

PROFESSIONAL SUMMARY

Data-focused engineer with 10+ years of experience in analytics, systems engineering, and data workflow automation, with several years building and maintaining ETL pipelines that support cross-functional teams and business decision-making. Strong proficiency in Python and SQL.

TECHNICAL SKILLS

Data and Processing: Python, SQL, Dagster, Airflow, dbt, Pandas, DuckDB

Cloud & Infrastructure: GCP (GCS, BigQuery), AWS (S3, Athena, Glue), PostgreSQL, Docker, Terraform

PROFESSIONAL EXPERIENCE

Senior Systems Engineer

L3Harris Technologies, Inc

October 2015 – Present
Salt Lake City, UT | Remote

- Designed and automated end-to-end ETL pipelines using Python, SQL, and Dagster to ingest, validate, transform, and orchestrate data movement between analyst teams, replacing manual, UI-dependent workflows.
- Developed automated pipeline triggers and validation logic to detect new input datasets, verify steady-state conditions, and prevent redundant downstream processing.
- Modeled and transformed production datasets to support downstream analysis, improving data accessibility and reducing data preparation time by over 50%.
- Reconciled IDs between evolving schemas to automate data movement and eliminate redundant work across teams while logging noncritical mismatches for analyst review and resolution.
- Built batch ingestion workflows to migrate flat-file datasets into centralized databases, reducing manual processing time from minutes to seconds per record.
- Collaborated with cross-functional teams to define data requirements and support system-level data architecture and integration decisions.
- Led teams of analysts while mentoring junior engineers, coordinating technical efforts, and driving the successful delivery of customer-facing projects.

Intern, Advanced Engineering

Teleflex, Inc

May 2014 – August 2014
Reading, PA

- Developed a C-based prototype device for ingesting and processing real-time image data for catheter tracking.

SELECTED TECHNICAL EXPERIENCE

NYC Building Permits Mapping

- Built and orchestrated a batch ELT pipeline using Python, Airflow, dbt, Terraform, and Docker to ingest, load, and transform NYC building permit data into BigQuery for analytics.
- Developed an incremental dbt model to transform raw data in BigQuery and test the resulting output.
- Implemented partitioning, clustering, and chunked ingestion strategies to enable efficient querying and large-scale historical backfill processing.
- Built an AWS implementation using S3, Glue Data Catalog, Athena, dbt, Terraform, and Apache Iceberg, with incremental merges and automated data quality tests.

EDUCATION

Master of Computer Science

University of Illinois Urbana-Champaign

2023

Bachelor of Science in Electrical Engineering

Brigham Young University

2015